YIFEI SUN

Full Stack Software Engineer · Cross-Platform Systems & UI · 13 Years

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SUMMARY

Full-stack engineer with 13 years of experience shipping end-to-end features across backend services, web, and mobile (Android). Deep expertise in Java/Spring Boot, TypeScript/React, GraphQL federation (Netflix DGS), and distributed systems at scale. Track record of building cross-platform products from 0-to-1, leading multi-team modernizations, architecting high-traffic APIs and internal UI platforms, and driving A/B experimentation. Combines strong product instincts with GenAI fluency — recently built an MCP-compatible server for LLM integration and a hybrid GenAI decision engine. Proven mentor who elevates senior engineers toward Principal-level scope.

LEADERSHIP & TECHNICAL HIGHLIGHTS

- Shipped full-stack features end-to-end — backend APIs, web UI, and mobile — across multiple platforms
- Led multi-team platform modernizations involving 10+ engineering teams across 3 departments
- Built Netflix DGS subgraphs and GraphQL federation gateways serving cross-platform consumers
- Mentored senior engineers on system design, cross-platform architecture, and path to Principal scope

Technologies: Java, Spring Boot, GraphQL (Netflix DGS), TypeScript, React, Angular · Android (Java), Web · AWS, Kubernetes, Kafka, Redis, PostgreSQL, MySQL · GraphQL Federation, Microservices, Event-Driven, CQRS · OpenAI/ChatGPT, MCP Protocol · A/B Experimentation · Datadog, Grafana

INDEPENDENT PROJECT

GenAI Decision Engine & Experimentation Platform

- Architected a full-stack hybrid rules + multi-path GenAI engine (React UI + Java/Python backend) with a policy overlay over heterogeneous telemetry, producing explainable real-time recommendations.
- Designed a configurable policy framework with DB-backed enterprise configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions.
- Built "Prompt Lab" — an end-to-end experimentation UI for historical backfill, decision simulation, offline evaluation/backtesting, and prompt/parameter search (grid & Bayesian optimization) with multi-provider LLM support.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead Oct 2019 – Present

GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph for the Delivery Lookup service into CarGurus' federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to avoid JIT compilation latency spikes on cold starts.
- Built load tests and staging GraphQL integration tests; migrating from legacy REST endpoints to support the new AI chatbot/shopping assistant platform.

High-Traffic API Modernization & Internal UI Platform

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions); the new implementation handled 50% of traffic during A/B testing.
- Introduced a new orchestration service and GraphQL boundary, consolidating 12 downstream integrations into 5 well-defined services, reducing fan-out and simplifying failure modes.
- Built internal UI tooling and admin platforms (React/TypeScript) to support operations, configuration management, and real-time monitoring of the modernized services.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined architectural standards for new microservices; built a parity-testing framework validating full payload and data-integrity against the legacy stack; removed years of hidden domain logic with clear documentation.

ChatGPT Apps SDK MCP Server (0→1)

- Sole engineer designing and implementing an MCP server compatible with the OpenAI Apps SDK, exposing CarGurus capabilities into ChatGPT as a high-visibility proof-of-concept.
- Delivered a performant, end-to-end implementation to unblock demos for PMs and directors, iterating on ambiguous requirements; designed for safe, incremental production rollout alongside existing APIs and clients.

Digital Shopping & Online Leads Products (0→1)

- Led full-stack design and implementation for a new digital shopping product launched during the pandemic, building both the consumer-facing web experience and the backend microservice/event-driven architecture for inventory, merchandising, and lead flows.
- Subsequently led development for additional digital commerce experiments, enabling the company to quickly launch and safely retire product bets via A/B experimentation without destabilizing core systems.

Cross-Org Platform & API Leadership

- Established patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored key design docs, RFCs, and decision records that drove consensus across platform and product teams.
- Mentored multiple senior engineers on system design, cross-platform architecture, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization (Full Stack)

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, building both the backend integration layer (Java) and the frontend login/identity flows (Angular 1 & 2, TypeScript) across internal systems, an enterprise partner platform, and a large external community.
- Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration & Internal Tools

- Built internal workflow tools (TypeScript/Angular) and partnered with product, support, and customer-facing teams to modernize enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Full-Stack Product Development — Web & Android

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core features across both the Android POS application and the web management dashboard, along with foundational backend services (Order Service, Labor Service, APIs).
- Shipped cross-platform features end-to-end — from backend API design to Android tablet UI and web admin interfaces — enabling restaurant operators to manage orders, labor, and reporting across devices.

Platform Infrastructure & Quality

- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional data sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010